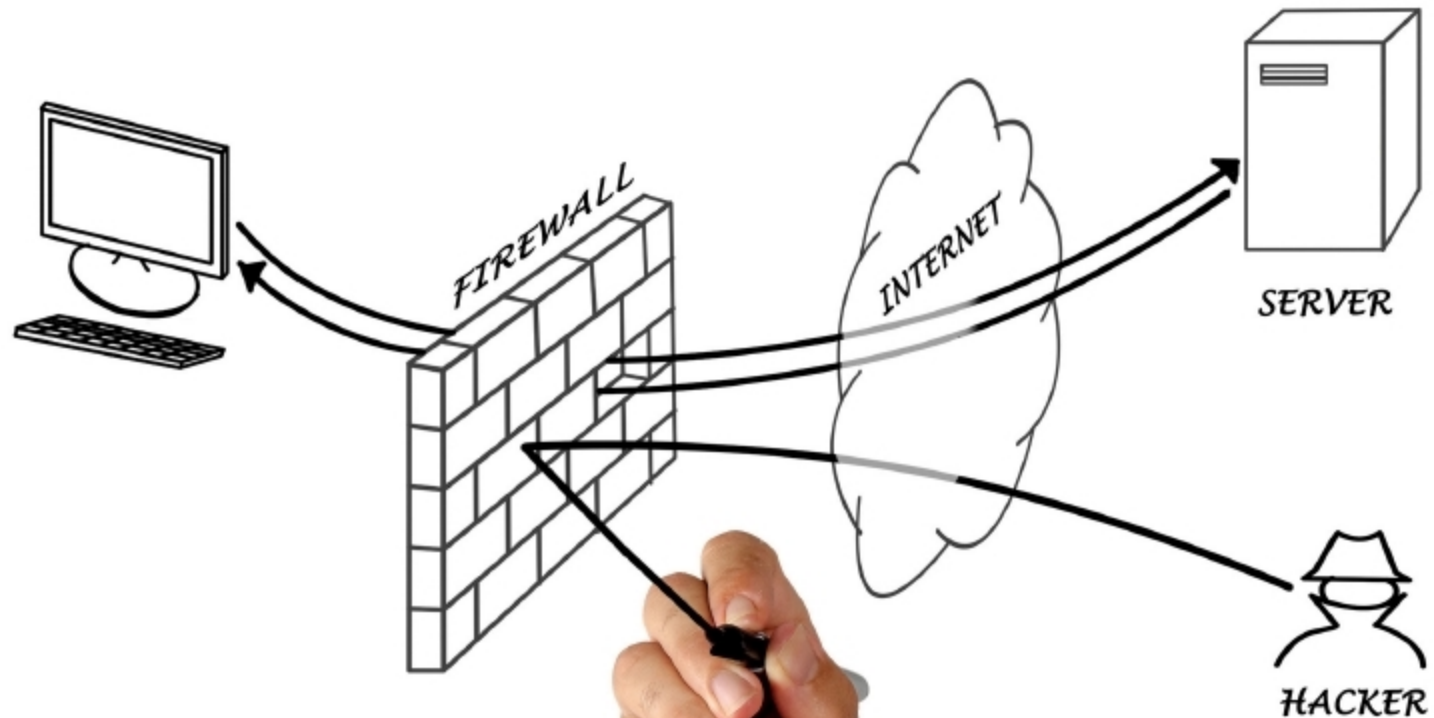


An Instant Guide to the Shorewall Firewall

Firewalls have gained importance due to the increasing number of malicious attacks on networks and computers. Shorewall, a powerful offering from the Linux world, is primarily used to protect networks, as its strength lies in working with zones.



A firewall provides an additional layer of defence, insulating the internal systems from external networks. The primary task of a firewall is to examine all the traffic routed between two or more networks according to certain defined rules and policies. Only if the terms of the rules and policies are complied with, is traffic routed between the networks; otherwise, all the packets are dropped. There are two access denial methodologies followed by firewalls, i.e., *allow traffic* or *deny/restrict traffic*. A firewall may allow all traffic through because it meets specified criteria, or it may deny all the traffic unless it meets certain criteria. The type of criteria used to determine whether the traffic is allowed through depends on how the user configures the firewall settings. Firewalls can also use complex rules as the basis to analyse the application data, prior to determining if the traffic should be allowed or rejected.

A firewall plays a crucial role in preventing unauthorised access to devices or networks. Its function is to carefully inspect the data entering or exiting the device, based on user configurations and deny access to data that comes from a suspicious network. It can also be used to log all attempts to enter the private network and trigger alerts when any sort of unauthorised entry is attempted.

Types of firewalls

There are primarily two main types of firewalls – hardware firewalls and software firewalls.

Hardware firewalls: These are systems that are independent of the computers that they protect. They filter information from the Internet as it passes into a computer. Most of the Wi-Fi or Internet routers have inbuilt firewalls for packet filtering. In general terms, hardware firewalls work by examining the data that flows in from the Internet and verify whether that information is safe. Simple firewalls, known as packet filters, examine the data itself for information such as its location and its source. Then the information the firewall gathers is compared to a set list of permissions in order to determine whether the information should be dropped or allowed through.